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Identifying, prioritizing and ranking of factors influencing the PhD milestones' completion delays

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Abstract
The research oriented manpower is instrumental in the growth of economies. Unfortunately, the growth of doctoral research community is not keeping pace with the growing research requirements of India. "Time" to complete the milestones during the doctoral tenure is considered as an important constraint in doctoral degree completion. The present study mainly focus on this 'time' factor to complete the milestones in a doctoral life cycle. Secondary data and data collected through a focus group questionnaire survey were utilized. The opinions out of the survey were pairwise compared and ranked utilizing the Thurstone scale. The five doctoral lifecycle milestones identified through focus group discussions were enrolment, registration, synopsis submission, thesis submission, and viva-voce examination. The lack of administrative efficiency, guidance and publication; and delays in coursework completion and report arrival were found to be the most influential factors in delaying the different milestones completion in doctoral research. The findings out of this research can be further utilized to develop different policies towards overcoming these hurdles that can in turn lead to the timely completion of the different doctoral milestones. Moreover, this study can be generalized to other academic organizations as the doctoral journey milestones remain similar in every academic system in the world. Finally, some useful policy insights out of this study are presented.

Keywords Doctoral degree completion delay · Doctoral life cycle milestones · Paired comparisons · Preference proportion matrix · Coefficient of concordance · Thurstone scale

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1 Introduction

Higher education, specifically at the doctoral level, plays a pivotal role in creating a knowledge society in the context of India. Competitive knowledge leaders with innovative minds are going to be the key for success in today's growing and knowledge transforming economy (Sahoo et al. 2017). Academic institutions play a major role in this regard. Although India has a large pool of manpower with undergraduate and graduate degrees in different disciplines, the numbers of research degree recipients are still rather small compared to other similar countries in the world (Kurup and Arora 2010; Menon et al. 2018) (Fig. 1). In the past decade there has been an increased awareness in the academic institutes of India towards the need of increasing doctoral graduates (PhD) number. With the advent of globalization and subsequent events, the stakeholders that include government as well as industry, are looking for increased number of doctoral degree holders to cater to the new challenges. In the last two decades, a number of technical and non-technical colleges and universities were established in India. These educational institutions are in public, private, or public–private partnerships. These academic institutions have been producing graduate and post graduate students in the last two decades. However, this is not sufficient to prepare intellectual leaders who will do the groundwork to create new paradigms and products for tomorrow. A sufficient as well as vibrant doctorates' pool, on the other hand, can contribute to the continuing economic growth of the country (Menon et al. 2018).

The Indian Institutes of Technologies (hereafter IITs) along with the Indian Institute of Science (IISc) are among the best academic institutes in India. The Kakodkar committee appointed by the Ministry of Human Resource Development (hereafter MHRD) of the government of India recommended autonomy measures to facilitate the IITs to scale up the doctoral population in these institutes (Kakodkar et al. 2011). The major concern of this committee was to show a path to improve the research scenario in the IITs by removing any of the constraints and attracting more candidates to join the doctoral program. As such, there are several constraints being cited as the hurdles in completing the doctoral research (Manathunga 2005; Paliktzoglou and Suhonen 2011; Reddy et al. 2016; Watts 2008).

One of the main hurdles in the doctoral program completion is the 'completion time', where the time management has emerged as an important factor in doctoral degree completion process (Araújo 2005; Harman 2002; Kolmos et al. 2008; Manathunga 2005;

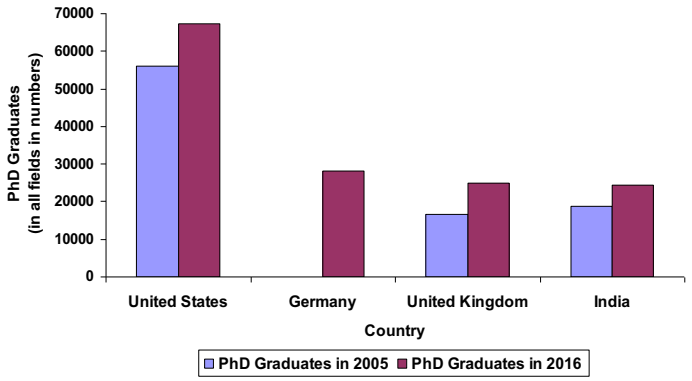


Fig. 1 Comparison of annual Ph.D. degree recipients of United States, Germany, United Kingdom and India (2005 data for Germany is unavailable). (*Source* Kurup and Arora 2010; OECD 2016)

Mowbray and Halse 2010; Paliktzoglou and Suhonen 2011; Watts 2008). The completion time has a significant impact in the life of a student as well as his/her supervisor (faculty). It is expected that a student joins a PhD program with a certain future aim in his life. The delay in completing the degree may put him behind the race. Additionally, the stipend/scholarship/assistantship given to a PhD student is also time dependent. A student can get the government scholarship maximum for a period of 5 years in the Indian context. Similarly, the students who have joined through 'projects' get their stipend from the project fund only. Once the project is over, the scholar will have to continue their doctoral research without receiving any stipend further. Many sponsored candidates join the PhD program with a certain time/money bond from their parent organizations. A delay in completing the degree will in turn push them to 'stress'. Furthermore a faculty's promotion in Indian universities are highly dependent upon the number of PhDs (degree awarded) he/she have guided. Accordingly, PhD completion delay of his/her students may also restrict them from further academic promotions. Hence PhD completion delays do have much influence over the future prospects of both the faculty and the scholar. In this backdrop, this paper aims to (1) identify the doctoral lifecycle milestones, (2) identify the factors that might influence these doctoral milestones, and (3) prioritize and rank these factors in order of their influence in delaying the identified doctoral milestones. The 'milestones' and their associated delaying factors in doctoral journey are identified and discussed in the literature review section. The data is collected through focus group discussions and a questionnaire survey. The possible time delaying factors are listed and ranked from the doctoral supervisors' perspective. These time delaying factors identified through the questionnaire survey are pairwise compared and are ranked utilizing the Thurstone scale for comparative judgments (Jones and Alcock 2014) in order of their importance.

2 Literature review

Higher education, specifically doctoral education, holds much importance in today's globalized knowledge society. Doctoral research plays an important role in driving innovation, growth, knowledge generation and research outputs for a country (Smith et al. 2010). This paper discusses the following issues in this regard in the subsequent sub-sections: (a) current scenario of doctoral degree completion across the globe, (b) variations in time of doctoral degree completion, and (c) doctoral life cycle process and time factor involved in Indian institutes. The doctoral life cycle research issues are delineated thereafter.

2.1 Current scenario of doctoral degree completion across the globe

There exists several studies that show the variation in doctoral completion times across different disciplines of research (Bair et al. 2005; Kim and Otts 2010; Sowell et al. 2015). Sowell et al. (2008) identified that the 10 year doctoral completion rates are 64% for engineering, 63% for life sciences, and 56% for social sciences. Fuller (2015) found that it took around 7 years to complete the doctoral program by 70 to 87% of the full-time doctoral candidates across renowned UK institutes. Palmer (2015) presented detailed statistics of 10 Australian universities with completion ratios of 62 to 72%.

The statistics related to Russian government shows that only 25 to 30% students have defended their doctoral thesis during the scheduled completion time. Vidak et al. (2017) identified insufficient supervision, lack of proper PhD structure, and institutional support

as the key factors for long duration of doctoral studies. Geven et al. (2018) suggested that a proper structural transition of doctoral degree might affect the PhD completion rate. Spronken-Smith et al. (2018) explored the factors that may affect the PhD completion rate with respect to the data from a New-Zealand based university. This study identified quality of candidate, high-quality supervision, funding facility etc. as some of the key factors for successful PhD completion. Zhou and Okahana (2019) have explored the influence of departmental support on the doctoral degree completion rate and time utilizing the data from 212 USA universities. Moreover, Maloshonok and Terentev (2019) have investigated about the national level barriers that may impact the doctoral completion rate for Russian educational institutes through an empirical study.

2.2 Variations in time of doctoral degree completion

To remain competitive in the global economy, it is essential for nations to enhance the numbers of graduate, post graduate and doctoral students and create conditions for timely completion of the degrees. With time, there are now less government funds, longer time requirements for doctoral degree, and higher institutional competition for students (George 2006). Taylor and Beasley (2005) reported that only 50% of the doctoral students are able to complete their degree requirements and, out of them, only a small fraction does so in the specified time frame. Moreover, the problems with timely completion of doctoral degree are well documented around the world (Green and Usher 2003; Neumann 2005; Rodwell and Neumann 2008).

In a study on the doctoral candidates from the Netherlands, Schoot et al. (2013) observed that the delay in PhD completion can lead to attrition/termination and hence can be detrimental to the university. The PhD completion rate or time to doctoral degree completion depends on a number of institutional factors including quality of supervision and quality of the candidate. Kyvik and Olsen (2014) found that PhD non-completion rate is very high in Norway. Motseke (2016) investigated the reasons for long time requirements for the adult learners to complete doctoral degrees with the help of data collected from a South African township through a questionnaire and focus group meeting. It was found that lack of quality, lack of facilities such as access to computers, supervision problems as well as employers' workload contributed to the long completion time. Geven et al. (2018) observed that in American schools, the median time for a doctoral degree completion is around 7.5 years after enrollment, and that this number has decreased marginally from 8.7 years since the year 1988. Horta et al. (2019) explored the effect of funding on the timely completion of doctoral degrees through a USA based single university case study. The study found that the unfunded PhD students, even with higher productivity, took longer time to complete; whereas funded PhD students with published papers took lesser time to complete. Based on a study in Portugal, Horta and Santos (2016) showed that the doctoral students, who publish during their PhD tenure, have greater research productivity throughout their career compared to those candidates who did not publish their research. Baker and Pifer (2015) found that person-environment fit, person-culture fit, and person-vocation fit provide critical insights into the experience of a doctoral student thereby offering a framework based on antecedents and outcomes to support future research. This work was extended by Ward and Brennan (2018) by developing a framework that provided insights on the association between the fit and the performance of doctoral students.

2.3 Doctoral life cycle process and time factor involved in Indian institutes

Unlike other degrees, the doctoral degree is not time dependent, rather it is event dependent. It is a collection of sequencial events. It starts from joining the programe and ends with successfully delivering the final viva-voce. Discussion with a number of faculty members and students, and Ph.D regulations available in the websites of different Indian academic institutes (IITB 2020; IIT Bhubaneswar 2020; IIT KGP 2020) aided this paper in conceptualizing the doctoral programe in the context of India. In general, once a doctoral student joins an Institute in India, he/she is assigned under a mentor or supervisor, followed by a committee called research scrutiny committee or doctoral scrutiny committee. This committee assign a set of coursework to the student based on his/her research area in a broader aspect. This whole procedure is normally called 'Enrollment'. Once the student completes the assigned coursework, he/she has to clear a comprehensive viva examination. There after the student proceeds for his/her research domain. Once the research is properly defined, the student is asked to present a seminar, normally called 'Registration' or 'registration seminar', in front of the scrutiny committee. After successfully delivering the registration seminar, the student conducts the research. Once the student completes the research, another seminar has to be delivered infront of the scrutiny committee called the 'synopsis seminar'. Once it is accepted by the committee, the student starts writing thesis that needs to be submitted to the parent Institute for evaluation by the external examiners. Normally, there are two examiners to evaluate one thesis. Once they agreed with the thesis findings, the student is asked to deliver a final seminar called 'viva-voce seminar' which may be in front of the open audience. Even one of the external examiners will also be present in person during the seminar, or through web mode. The successful delivery of the viva-voce seminar defines the successful completion of the doctoral journey.

Normally in Indian context, a student gets the government scholarship for a period of 5 years or up to submitting the thesis, whichever is earlier. Accordingly, the student has to complete all the above steps within a time period of five years. It is found that on an avaerage a student should complete the coursework and comprehensive examination within one year, should deliver the registration seminar within 2 years, should deliver the synopsis seminar within 4.5 to 4.75 years and should submit the thesis within 5 years. In normal scenario, the final viva-voce is not dependent on the student directly, rather the administrative efficiency of the Institute, response from the examiner etc. are found to be key factors on this aspect.

Apart from the doctoral students, who get 5 years of scholarship from the Government, there are other categories of scholars like sponsored Ph.D. (through some sponsoring agencies), self-sponsored or individual Ph.D etc. Usually these students have to carry out their doctoral programe from their professional organization where they do not get the sufficient amount of academic facilities that are needed for a doctoral student. Hence, it is found to be dificult for them to complete their doctoral degree within the stipulated time. Even many of the government sponsored students also cannot complete their degree within the stipulated time period which is attributed to factors like the lack of publication, compatibility mismatch with the supervisor etc. All these issues signifies that there are factors that can influence the completion time of doctoral degree in a direct or indirect manner. Thus, the time management issue has emerged as an important factor in the doctoral degree completion process (Araújo 2005; Harman 2002; Kolmos et al. 2008; Manathunga 2005; Mowbray and Halse 2010; Paliktzoglou and Suhonen 2011; Watts 2008).

2.4 Doctoral life cycle research issues

It is important to understand the constraint factors throughout the doctoral education journey of a student as it is of benefit to the doctoral educators striving to improve the doctoral completion rates (Ward and Brennan 2018). The doctoral degree can be called as event-driven unlike the under-graduate or the post-graduate degrees, which are mostly time bound. There exist literature that have shown the variations in PhD completion rates with respect to different countries or different research areas, and minimizing of the doctoral completion time as a key research issue (Schoot et al. 2013; Motseke 2016; Geven et al. 2018). The key factors that are responsible for the PhD completion delay are also discussed in the literature. Some of these includes studies on factors of departmental or institutional support (Lin and Chiu 2014; Motseke 2016;Vidak et al. 2017; Wamala et al. 2011; Zhou and Okahana 2019), support from the supervisor (Spronken-Smith et al. 2018; Maloshonok and Terentev 2019; Vidak et al. 2017; Motseke 2016; Schoot et al. 2013), publication (Horta et al. 2019), fund support (Spronken-Smith et al. 2018), institutional factors (Geven et al. 2018; Motseke 2016; Main 2014 2018; Schoot et al. 2013), proper PhD structure (Vidak et al. 2017), and fulfillment of required criteria (Main 2014). However, the studies on ranking these factors based on their importance towards the timely completion of doctoral degree specifically in the Indian context are sparse. Additionally, ranking those factors in order of importance for different milestones in a doctoral life cycle is also not widely researched in the literature. In this connection, need exists to divide the doctoral life cycle into different milestones and then rank different factors that may influence the timely completion of these doctoral milestones in the Indian context.

3 Research framework adopted

The primary sequence of steps followed by this study are: (1) Conceptualizing the doctoral life cycle and categorizing the doctoral students into science and engineering discipline, (2) Identifying the key milestones in a doctoral life cycle, (3) Identifying the key factors that may influence the time to complete these milestones for a doctoral student, and (4) Ranking those factors according to their order of influence for different milestones of different categories of doctoral life cycles. The framework of the present study is outlined in Fig. 2.

Initially, the doctoral life cycle milestones and all the possible factors expected to influence these milestones are identified through literature survey and focus group discussions (Sects. 4 and 5). Then a questionnaire survey (Sect. 6) is utilized to identify those specific factors from the total pool that are to influence each of these doctoral life-cycle milestones. The factors are pairwise compared using Thurstone scale. The basic steps followed by the present procedure are as follows.

Step 1 Each of the factors for a given milestone (i.e., Time to Registration, Time to Synopsis etc.) are pairwise compared with each other, and the results are given in percent terms.

Step 2 The corresponding 'Z' values for each of these percent values are calculated that is followed by the calculation of mean 'Z' value and total 'Z' value for each of the column.

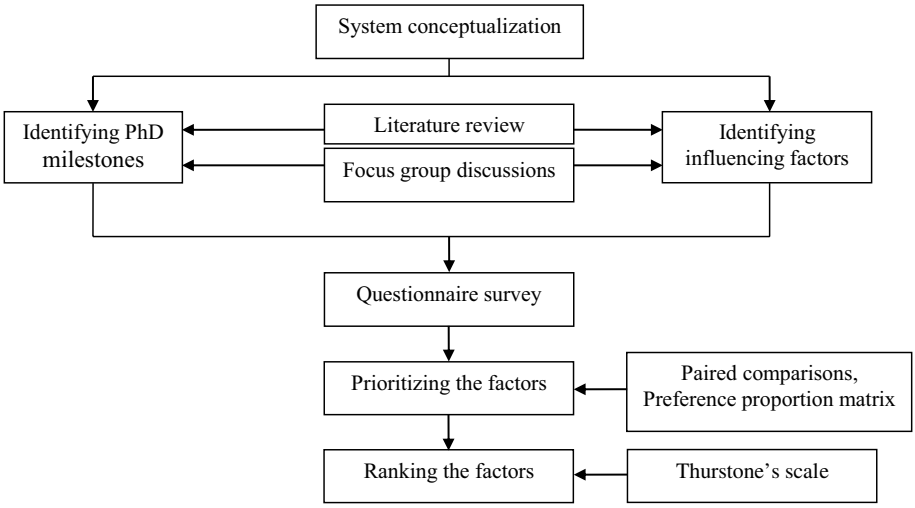


Fig. 2 Research framework adopted

Step 3 The adjusted *R* values are obtained by subtracting the minimum among the Mean (Z) to all the Mean (Z values (including itself).

Step 4 These adjusted *R* values are then used to draw the Thurstone scale.

Thus, the questionnaire survey data is utilized (Sect. 7) to develop the preference proportion matrix which was further used to develop the Z-values and adjusted R values. These adjusted R values are finally transformed to a Thurstone scale for ranking the factors that influence the time taken in completing the different milestones in the doctoral life cycle.

4 Identifying the milestones in doctoral journey in Indian context

A doctoral degree is a sequential collection of stages or milestones. A student is not allowed to go to the next event until he/she completes the previous milestone. So a doctoral student has to go through a number of key milestones before completing the degree. Thus, focus group discussions were conducted to identify the key doctoral milestones that may get delayed due to some external factors. Nyumba et al. (2017) defines a focus group discussion as "a qualitative approach to gain an in-depth understanding of social issues. The method aims to obtain data from a purposely selected group of individuals rather than from a statistically representative sample of a broader population". The focus group discussions were carried out among the academicians from a reputed research-oriented Indian academic institute. The participants included assistant professors, associate professors and professors of the institute and all of who have guided and presently guiding students towards their doctoral degrees. Hence the focus group members are highly experienced and knowledgeable in the doctoral guiding process. The focus group discussions concluded that the doctoral students in India mainly go through five key milestones in their entire doctoral journey. These five key milestones include (1) enrolment, (2) registration, (3) synopsis submission, (4) thesis submission, and (5) viva-voce examination. The enrolment stage

involves admitting a candidate to the doctoral degree program which also includes the allocation of course work and formation of the doctoral scrutiny committee. The registration involves completion of all the assigned course works, and defining and presenting the research problem and research objectives thereby officially formalizing the research topic. The synopsis submission is the pre-final stage involving presenting and submitting the findings and conclusions out of the research activity by the doctoral candidate. This is followed by the official submission of the thesis for expert review. Finally the student appears for the viva-voce/defense seminar that realizes the doctoral degree completion. The focus group discussions further help to combine, filter or rename the factors (mentioned in Sect. 2.4) that may be responsible for the delays in completion of the different milestones in a doctoral life cycle in Indian context. The details about these factors are given in Sect. 5.

5 Identifying the factors affecting the doctoral milestones' completion delay

The literature discussed under Sect. 2.4 aided in identifying some of the key factors that are expected to cause the delays in completion of the doctoral life cycle milestones. The focus group meetings were further conducted to combine, filter or rename these factors (identified from the literature study) that cause the doctoral milestones' completion delays. The output from the literature study and the focus group discussions identified the following 8 factors: lack of administrative efficiency (AE; also identified by Wamala 2011; Zhou and Okahana 2019), lack of academic facility (AF), delay in department (DD; also identified by Wamala et al. 2011; Zhou and Okahana 2019), not fulfilling required criteria (FRC; also identified by Main 2014), delay in coursework completion (CC), lack of guidance (LG; also identified by Maloshonok and Terentev 2019; Schoot et al. 2013), lack of publication (PB; also identified by Horta et al. 2019), and thesis report arrival delay (RAD) that are expected to be the influential factors in PhD milestones' completion delays. The descriptions of each of these factors are given in the "Appendix 1". The participants of the questionnaire survey were asked to rank these identified factors in order of importance that cause delays in the doctoral life cycle.

6 Questionnaire survey

The questionnaire survey was conducted among the faculty members of a premiere Indian academic institution. A questionnaire survey is a research instrument used in a study to gather primary data in a valid, reliable and unbiased manner. A questionnaire survey is a good way, often the only way, of getting a picture of the current state of a group (Janes 1999).

A structured questionnaire has been designed to analyze the 'time' variable in completing the doctoral degree process due to the impact of the 8 factors identified in the present study. A total of 67 responses were collected from the faculty of engineering, and the science and humanities departments. A summary table ("Appendix Table 6") for questionnaire responses is given in "Appendix 2". The respondents were asked to select 4 out of the 8 factors that they perceive to be responsible for the delays in completion of each of the PhD milestones already defined. The questionnaire survey responses were categorized in to three sections of (i) "all disciplines section" which includes responses of all the

faculty members, (ii) "engineering section" which includes responses of the faculty members of engineering departments, and (iii) "science & humanities section" which includes responses of the faculty members of science and humanities departments.

7 Data analysis

The questionnaire administered was for identifying the most relevant 4 out of the 8 factors that are influencing each of the five PhD life cycle milestones. Initially, the frequencies of occurrence of each of the 4 factors given by each of the 67 respondents were computed for the five doctoral milestones. Finally, 4 factors based on the highest frequencies of occurrence for each of the milestones were selected (Table 1).

Table 1 shows the top 4 prioritized factors (out of 8 factors) that are influencing the PhD milestones' completion delays. However, the priority order of influence of these factors over each of the milestones in delaying the PhD completion is still unknown. Paired comparisons based relative ranking method is utilized to obtain the most influential Thurstone's Case V model of comparative judgment (Maydeu-Olivares 2004) to rank these top 4 factors for each of the milestones. Thurstone scale is widely utilized for comparative judgement studies and analysis. The application of Thurstone scale in academic/education system studies can be found in Arfandi et al. (2020), Moxham and Moxham (2007) and Wheadon et al. (2020). The software Minitab is utilized for the computation of the same. The development of Thurstone scale for the five milestones in the doctoral life cycle is computed in the following sub-sections.

7.1 Enrollment

In the questionnaire, the top 4 factors that are found to influence the delay in the 'time to enrollment' are lack of administrative efficiency (AE), lack of academic facility (AF), delay in department (DD), and not fulfilling required criteria (FRC). The first step involves the computation of a preference proportion matrix obtained from the responses. A sample of the calculation for 'all disciplines" category is given in Table 2. Now, based on Table 2, Table 3 is formed which summarizes the Z-values (area under a normal curve that a given proportion occupies) appropriate for each preference proportion.

Table 1 Top 4 factors for each of the doctoral milestones listed by the respondents

Doctoral milestones	Top 4 factors
Enrollment	AE, AF, DD, FRC
Registration	AE, AF, CC, LG
Synopsis submission	AE, AF, PB, LG
Thesis submission	AE, AF, PB, LG
Viva-voce	AE, AF, RAD, LG

AE: administrative efficiency, *AF:* lack of academic facility, *DD:* delay in department, *FRC:* not fulfilling required criteria, *CC:* delay in coursework completion, *LG:* lack of guidance, *PB*: lack of publication, *RAD:* thesis report arrival delay

Table 2 The preference proportion matrix (in percentage) for enrollment (All disciplines)

Factors	Preferred factors			
	Lack of administrative efficiency	Lack of academic facility	Delay at department	Not fulfillment required criteria
Lack of administrative efficiency	0	17.91	22.39	80.3
Lack of academic facility	82.09	0	74.24	91.04
Delay at department	77.61	25.76	0	85.07
Not fulfilling required criteria	19.7	8.96	14.93	0

The adjusted *R* (R-adj) values are obtained by subtracting the minimum of Mean Z-values from itself thereby normalizing that value to '0' and affecting the same operation to all the remaining Mean Z-values (Table 3). This is done to obtain the reference point or origin of the interval scale. The R-adj values give an interval scale along which the stages can be placed. Based on the R-adj values the Thurstone scale is developed (Fig. 3).

Similarly the Thurstone scale was computed for engineering discipline, and science & humanities disciplines (see Fig. 4 below). The extent of agreement among the survey respondents with respect to the final outcome is also measured using the Kendall's *W*-statistic known as Coefficient of Concordance (*u*) (for application of Kendall's Coefficient of Concordance see Field (2014); Gearhart et al. (2013); Grzegorzewski (2006); Li et al. (2020)). The coefficient of concordance is the measurement of agreement among the respondents with respect to the final outcome (i.e. Thurstone scale) and is measured through the coefficient of concordance (agreement) for different milestones of different doctoral students' categories. To calculate this, another preference

Table 3 Z-values related to preference proportion matrix (in percentage)

Factors	Lack of administrative efficiency	Lack of academic facility	Delay at department	Not fulfillment required criteria
Lack of administrative efficiency	0.00	−0.92	−0.76	0.85
Lack of academic facility	0.92	0.00	0.65	1.35
Delay at department	0.76	−0.65	0	1.04
Not fulfilling required criteria	−0.85	−1.35	−1.04	0.00
Total	0.83	−2.92	−1.15	3.24
Mean	0.2075	−0.7300	−0.2875	0.8100
R-adj	0.9375	0.0000	0.4425	1.5400

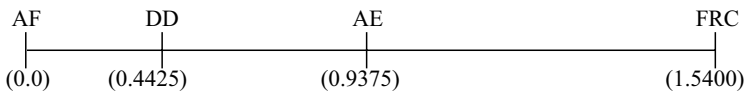


Fig. 3 Thurstone's scale for all discipline category for 'enrollment'

Discipline	Relative Scale	*u*
Engineering	AF (0.0) DD (0.3200) AE (0.8125) FRC (1.9675)	0.50
Science & humanities	AF (0.0) DD (0.5525) AE (0.9350) FRC (1.3425)	0.34

Fig. 4 Thurstone's scale for engineering, and science & humanities category for 'enrollment'

matrix is formed. The only difference is that the entry in the Table 4 is integer instead of percentage.

$$Coefficient\ of\ concordance,\ u = \frac{4 \times (squared\ entry\ sum - sum\ of\ all\ entries)}{[m(m-1)\,n(n-1)] - 1}$$

where $m = number\ of\ respondents = 67$; $n = number\ of\ objects = 4$

Using similar procedure the mean, R-adj values and 'u' values for different categories (i.e. all disciplines, engineering discipline, and science & humanities discipline) for different milestones are computed further.

7.2 Registration

In the questionnaire, the top 4 factors that are found to influence the delay in 'time to registration' are lack of administrative efficiency (AE), lack of academic facility (AF), lack of guidance (LG), and delay in coursework completion (CC). Based on the data on the above four factors, the Thurstone scale is developed for all disciplines, engineering, and science & humanities categories (Fig. 5).

Table 4 Preference matrix for coefficient of concordance (u) for all discipline category for 'enrollment'

Factors	Preferred factors			
	Lack of administrative efficiency	Lack of academic facility	Delay at department	Not fulfillment required criteria
Lack of administrative efficiency	0	55	52	13
Lack of academic facility	12	0	17	6
Delay at department	15	49	0	10
Not fulfilling required criteria	53	61	57	0

Coefficient of concordance for all disciplines category for 'enrollment', $u = 0.2147$

Discipline	Relative Scale	*u*
All disciplines	AE (0.0) AF (0.8225) LG (1.3750) CC (1.4875)	0.44
Engineering	AE (0.0) AF (0.9800) LG (1.3400) CC (2.4350)	0.53
Science & humanities	AE (0.0) AF (0.6800) LG (0.9900) CC (1.8700)	0.45

Fig. 5 Thurstone's scale for all discipline, engineering, and science & humanities for 'registration'

7.3 Synopsis submission

In the questionnaire survey, the top 4 factors that are found for the delay in 'time to synopsis submission' are lack of administrative efficiency (AE), lack of academic facility (AF), lack of publication (PB), and lack of guidance (LG). Based on the collected data, the relative weight and the ranking among these 4 factors were computed and were represented on Thurstone Scale for all disciplines, engineering, and science & humanities categories (Fig. 6).

7.4 Thesis submission

In the questionnaire, the top 4 factors that are found to influence the delay in 'time to thesis submission' are lack of administrative efficiency (AE), lack of academic facility (AF), lack of publication (PB), and lack of guidance (LG). Utilizing the collected data,

Discipline	Relative Scale	*u*
All disciplines	AE (0.0) AF (0.96) LG (1.1725) PB (1.5875)	0.39
Engineering	AE (0.0) AF (0.96) LG (1.37) PB (1.5975)	0.38
Science & humanities	AE (0.0) AF (1.1525) LG (1.2275) PB (1.7900)	0.46

Fig. 6 Thurstone's scale for all discipline, engineering, and science & humanities for 'synopsis'

Discipline	Relative Scale	*u*
All disciplines	AE (0.0) — AF (0.7750) — LG (1.1000) — PB (1.3950)	0.34
Engineering	AE (0.0) — AF (0.6425) — LG (0.7425) — PB (1.0700)	0.46
Science & humanities	AE (0.0) — AF (0.7775) — LG (1.1775) — PB (1.5350)	0.42

Fig. 7 Thurstone's scale for all discipline, engineering, and science & humanities for 'thesis submission'

Discipline	Relative Scale	*u*
All disciplines	LG (0.0) — AF (0.4125) — AE (1.2850) — RAD (2.4525)	0.55
Engineering	AF(0.0) — LG(0.0250) — AE (2.5425) — RAD (3.5950)	0.61
Science & humanities	LG (0.0) — AF (0.4325) — AE (0.9525) — RAD (1.8125)	0.58

Fig. 8 Thurstone's scale for all discipline, engineering, and science & humanities for 'viva-voce'

the relative weight and the ranking among the 4 identified factors were found out based on Thurstone scale ranking method (Fig. 7).

7.5 Viva-voce

In the questionnaire, the top 4 factors that are found to influence the delay in 'time to viva-voce completion' are lack of administrative efficiency (AE), lack of academic facility (AF), thesis report arrival delay (RAD), and lack of guidance (LG). Utilizing the collected data, the relative weight and the ranking among the 4 identified factors were found out based on Thurstone scale ranking method (Fig. 8).

8 Results and discussion

The Table 5 summarizes the outcomes of the Thurstone scale analysis for different doctoral milestones in three categories. For the 'enrollment' phase, the most important ranked factor was 'not fulfilling required criteria' for all disciplines, engineering, and science & humanities categories. The least influential factor in completing

Table 5 Discipline wise ranking of factors for different doctoral degree milestones

Milestones	Influencing factors in order of importance				
	Disciplines	Rank 1	Rank 2	Rank 3	Rank 4
Enrollment	All disciplines	Not fulfilling required criteria	Lack of administrative efficiency	Delay at department	Lack of academic facility
	Engineering	Not fulfilling required criteria	Lack of administrative efficiency	Delay at department	Lack of academic facility
	Science and humanities	Not fulfilling required criteria	Lack of administrative efficiency	Delay at department	Lack of academic facility
Registration	All disciplines	Delay in coursework completion	Lack of guidance	Lack of academic facility	Lack of administrative efficiency
	Engineering	Delay in coursework completion	Lack of guidance	Lack of academic facility	Lack of administrative efficiency
	Science and humanities	Delay in coursework completion	Lack of guidance	Lack of academic facility	Lack of administrative efficiency
Synopsis submission	All disciplines	Lack of publication	Lack of guidance	Lack of academic facility	Lack of administrative efficiency
	Engineering	Lack of publication	Lack of guidance	Lack of academic facility	Lack of administrative efficiency
	Science and humanities	Lack of publication	Lack of guidance	Lack of academic facility	Lack of administrative efficiency
Thesis submission	All disciplines	Lack of publication	Lack of guidance	Lack of academic facility	Lack of administrative efficiency
	Engineering	Lack of publication	Lack of guidance	Lack of academic facility	Lack of Administrative Efficiency
	Science and humanities	Lack of publication	Lack of guidance	Lack of academic facility	Lack of administrative efficiency
Viva-voce	All disciplines	Report arrival delay	Lack of administrative efficiency	Lack of academic facility	Lack of guidance
	Engineering	Report arrival delay	Lack of administrative efficiency	Lack of guidance	Lack of academic facility
	Science and humanities	Report arrival delay	Lack of administrative efficiency	Lack of academic facility	Lack of guidance

the enrollment process is found to be 'lack of academic facility' in the Institute for all the three disciplines. For the 'registration' phase, the most influential factor in timely completion of registration/registration seminar was found to be 'delay in coursework completion', and the least influential to be 'lack of administrative efficiency' for all the three discipline categories. In case of the 'synopsis submission' phase and the 'thesis submission' phase, the most influential factor in timely completion of synopsis/synopsis seminar and thesis submission was found to be the 'lack of publications', and the least influential to be 'lack of administrative efficiency' for all the three discipline categories. Finally the successful timely completion of doctoral 'viva-voce' was highly influenced by 'report arrival delay' for all the three disciplines. In case of all disciplines and the science & humanities, the timely completion of 'viva-voce' was least influenced by 'lack of guidance', while it was found to be the 'lack of academic facility' for the engineering disciplines.

9 Practical and theoretical implications of the study

PhD is a sequential collection of milestones or events. The present study has conceptualized the doctoral life cycle process and divides it into the following sequence of milestones: Enrollment, Registration, Synopsis submission, Thesis submission and Viva-voce. This paper has discussed the significance of PhD life cycle delay in Sect. 1. As the PhD life cycle milestones are sequential in nature, delay in one milestone will restrict the next one to occur, i.e. a student cannot give registration seminar until his enrollment is complete, or synopsis cannot be delivered until registration seminar is over and so on. Some of the identified factors that cause these delays are related to PhD students (e.g. completing the coursework on time to be eligible for registration, fulfilling the required criteria to complete the enrollment, publish research articles to deliver synopsis or thesis etc.), some are associated to the faculties (e.g. proper guidance, help in publication to make the PhD students be eligible to deliver registration, synopsis and thesis on time), and even some are purely dependent upon the institute administration (e.g. transparency in displaying required criteria, administrative efficiency, academic facility etc.). As a whole, it shows that a proper coordination with full responsibility is required among the key stakeholders (PhD student, faculty and institute administration) involved in the doctoral program in completing the doctoral degree on time.

The questionnaire survey data was utilized for ranking the various factors that are to influence the delays in milestones in the doctorate degree completion journey. The key factors (with respect to the order of importance and the frequency of occurrences) that are found to be the main reason behind the delay in doctoral life cycle are coursework completion delay, publication, lack of guidance, administrative efficiency, report arrival delay etc. The past literature and the focus group discussions have identified these factors. The Thurstone scale was developed for each milestone thereby ranking the various influential factors in order of importance. The delay in requirement criteria fulfillment, coursework completion, lack of publication and report arrival delay were found to be the most influential factors (based on the ranking in Table 5) in delaying the different milestone completions in doctoral research. The factors of lack of administrative

efficiency and lack of guidance are also found to be relatively significant based on its frequency of occurrence in pair-wise comparison.

10 Conclusions, policy insights and future scope of study

The delay in doctoral degree completion time is one of the key issues addressed in educational research across the globe. The present study is on this 'time' factor involved in completing the entire doctoral program in an institute of national importance in India. The study considered engineering discipline, science & humanities disciplines, and both together under the category of 'all disciplines'. The present study has contributed with:

1. The understanding of the factors that lead to the delay in PhD milestones' completion in Indian context. As far the knowledge of the authors, the present study is the first of its kind in Indian context.
2. Prioritizing and ranking the factors in order of their influence in delaying the identified doctoral milestones. This can in turn help the policy makers to devise policies and action plans with greater focus on the ranked most influential factors.

In a nutshell, the results show that the time to enrollment can be minimized by promptly fulfilling the required criteria during joining the doctoral program, and registration delays can be reduced through timely coursework completion. Accordingly, the institute should inform the admitted doctoral students about the joining date, time, and mandatory documents to be submitted during joining the doctoral program. In parallel, the newly joined students should also be present in the admission desk on the joining day with the proper documents as instructed by the Institute to avoid the enrollment delay. In most of the Indian academic institutes, it is mandatory to complete a set of coursework before proceeding for the research. However, many a time, it is found that the students are not able to complete their coursework or comprehensive examination (related to their coursework) on time causing the registration delay to occur. The number of research publications is an important criterion for thesis submission. In practical life also, it has become almost mandatory for the doctoral students of all the premiere Indian academic institutions to have publications before delivering their synopsis or submitting thesis. Even, the UGC (University Grant Commission) has also made a clear guideline in this regard. As such by increasing the guidance efficiency, the research publication rates can be increased thereby reducing the time delays in delivering synopsis and thesis submission. Moreover the report arrival time can decrease the delays in viva-voce examinations. In majority of the cases, it is related to the external examiners and there is almost nothing to do from the student or the supervisor's side. However, from the Institute's part, one effective way to reduce the report arrival time is to follow-up with the thesis examiners at regular time intervals. This 'follow-up' can be realized through communicating over phones or by sending emails at regular time intervals to the external thesis examiners. Finally, the findings out of this research can be further utilized to develop different policies towards overcoming these hurdles that can in turn lead to the timely completion of the different doctoral milestones in Indian academic context. Moreover, the present study can be generalized to other academic institutes in the

world as the doctoral life cycle milestones remains more or less the same in all the academic systems.

The outcomes of the present study provide some insights on the ongoing policy of strengthening the research pool in Indian academia. The government policy of increasing the research outcome from the Indian academic institutes mainly involves scaling up of PhD candidates and which is carried out since the Kakodkar Committee Report (2011). Based on the study outcomes, the authors suggest that more focus shall be given to the timely completion of the doctoral program rather than on scaling up of the doctoral students' number. The delay in PhD completion will result in inducing stress and mental fatigue in both supervisor and the student. The PhD completion delays along with the PhD candidates scaling up policy can further increase this stress on the supervisor leading to more delaying of the PhD completions of the existing and the newly joined researchers. Moreover, the increased stress on the supervisor and the student will result in substandard research outcomes that may have no value addition to the society and the country. Hence, in light of the present research outcomes, the authors suggest that the primary focus shall be on the timely completion of the doctoral program. The outcomes of the present research can aid the policy makers in designing policies towards the timely completion of the doctoral program. The policy makers shall mainly focus on the presently prioritized and highly ranked factors that delay the doctoral completion milestones towards designing suitable policies to overcome these hurdles.

The scope of future work lies in further extending the present study by conducting similar questionnaire surveys among other relevant stakeholders such as doctoral students and administrative staff members of different academic institutes across India or across the globe. The milestones in the doctoral life cycle can differ in other countries. The present study considers limited number of broad level factors that are responsible for the delays in different doctoral milestones. However, there may be a number of micro-level factors which may more accurately define these doctoral delays. Studying and ranking those factors for different doctoral milestones need to be explored further.

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Availability of data and material Data will be made available on request.

Code availability No software codes were used in the present research.

Compliance with ethical standards

Conflict of interest There are no conflicts of interest/competing interests among authors.

Appendix 1

The meaning of the factors under the different PhD milestones and given in the questionnaire is as follows.

1. Lack of Administrative Efficiency (AE): This variable involves the delay in completing the required official formalities that need to be carried out by different administrative sections in the various stages of doctoral journey.
2. Lack of Academic Facility (AF): This variable shows the extend of availability of academic (i.e. computers, printers, books etc.) and research (i.e. labs, instruments, journals etc.) facilities in the academic institution.
3. Delay in Department (DD): This variable is on the delay from the academic department side in carrying out various formalities like conducting seminars, sending the reports for higher authority approval etc.
4. Not fulfilling Required Criteria (FRC): This variable is on fulfilling required criteria like minimum CGPA requirements, submission of educational and work-experience certificate and other credentials set by the institution for enrolling in the doctoral program.
5. Delay in Coursework Completion (CC): Coursework completion is a prerequisite for registering in the doctoral program. A delay in coursework completion will result in delaying the registration seminar.
6. Lack of Guidance (LG): This variable signals the extend of unavailability of the research supervisors to the doctoral scholars.
7. Lack of Publication (PB): The minimum number of research publications is a prerequisite in giving synopsis seminar and thesis submission in majority of the academic institutions in India. Hence the lack in required number of publications will delay the synopsis and thesis submission.
8. Thesis Report Arrival Delay (RAD): This is the delay in receiving back the evaluated thesis from external and internal thesis examiners. The delay in report arrival results in the delay in doctoral viva-voce examination.

Appendix 2

See Table 6.

Table 6 Summary table for questionnaire responses

Respondents' information	Sample number (Nos.)	Percentage (%)
Number of institute staff communicated	120	
Number of institute staff responded with the questionnaire	82	
Number of completed responses that are considered in the study	67	
Gender		
Male	38	56.72
Female	29	43.28
Discipline wise considered respondents		
Engineering	41	61.19
Science and humanities	26	38.81
Designation wise considered respondents		
Professor	22	32.84
Associate professor	26	38.81
Assistant professor	19	28.36

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